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Experiment No 3

Aim : create a database using DDL and apply integrity constraints.

Theory :

*) DDL commands :-

1. create Table command :- It defines each column of the table uniquely. Each column has minimum of three attributes, a name, datatype and size.

Syntax :

create table <table name> (<col> <datatype> (<size>), <col2> <datatype> (<size>))

Ex :- create table emp(
empno number(4) primary key,
ename char(10));

2. Modifying the structure of tables :

i) Add new columns

Syntax : alter table <tablename>
add (<new col> <datatype(size), <new col>
datatype(size));

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Ex:- alter table emp add (sal number(7, 2));

iii) dropping a column from a table

Syntax:- alter table <tablename>
drop column <col>

Ex:- alter table emp
drop column sal

iii) Modifying existing columns

Syntax:- Alter table <tablename>
drop column <col>

Ex:- OR

alter table <tablename>
alter column (<col> <newdatatype> (<newsize>)

Ex:- alter table emp
modify (ename varchar2(15));

iv) Renaming the tables

Syntax:- Rename <oldtables> to <newtables>

Ex:- rename emp to emp1

v) Destroying structure of tables

Syntax: drop table <tablename>

Eg:- drop table emp;

vi) Integrity constraints:-

constraints are the Business rules which are enforced on the data being stored in a table are called constraints.

Types of Data constraints

1. The Primary Key at column level

Syntax:-

```
CReate table tablename(
columnname1 datatype constraint <constraintname1>
 PRIMARY KEY,
columnname 2 datatype,
columnname 3 datatype)
:
)
```

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2. The primary key defined at table level

```
create table tablename(
columnname1 datatype,
columnname2 datatype,
columnname3 datatype,
Primary key (columnname1, columnname2));
```

3. The foreign key defined at column level

Syntax:

```
create table tablename(
columnname1 datatype references tablename[(columnname)]
[On delete cascade],
columnname2 datatype,
columnname3 datatype,
:
);
```

The table in which foreign key is defined is called foreign table or detail table

The table in which primary key is defined and referenced by foreign key is called primary table or Master table.

On delete cascade is set then delete operation in master table will trigger the delete operation for corresponding records in the detail table.

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4. The foreign key defined at table level

Syntax:-

```
create table tablename (
columnname1 datatype,
columnname2 datatype,
columnname3 datatype,
primary key (columnname1, columnname2),
foreign key (columnname2) references tablename2);
```

5. If constraint can be given user:-

defined name, the syntax is:
constraint <constraint name> <constraint definition>

6. The check constraint defined at column level:-

```
create table tablename (
columnname1 datatype check (logical expression),
columnname2 datatype,
columnname3 datatype,
...
)
```

7. The check constraint defined at table level:-

```
create table tablename (
columnname1 datatype,
```

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columnname2 datatype,
columnname3 datatype,
check (logical expression1),
check (logical expression2));

8. The Unique constraint defined at the column level:

create table tablename(
columnname1 datatype Unique,
columnname2 datatype unique,
columnname3 datatype .....);

9. The Unique constraint defined at the table level:

create table tablename(
columnname1 datatype,
columnname2 datatype,
columnname3 datatype,
Unique (columnname1));

10. Not Null constraint defined only at column level:

create table tablename(
columnname1 datatype Not Null,
columnname2 datatype Not Null,
columnname3 datatype,....);

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11. default constraint defined only to column level

columnname datatype default <value>

*) The Data Manipulation Language:

(DML) is used to retrieve, insert and modify database information.

Basic DML commands:

1. Insert
2. update
3. delete

1. Insert into:

This is used to add records into a relation

a) Inserting a single record

Syntax:

INSERT INTO <relation/tablename>
VALUES (data_1, data_2, ..., data_n)

Example:

INSERT INTO student
VALUES (1, 'Ravi', 'M Tech', 'Palakol')

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2. Update - set - where:

This is used to update the content of a record in a relation.

Syntax:

```
UPDATE relation name SET field name1 = data,
                         field name2 = data,
WHERE field name = data;
```

Example:

```
UPDATE student SET
sname = 'Kumar' WHERE Sno = 1;
```

3. Delete from:

This is used to delete all the records of a relation but it will retain the structure of that relation.

a) Delete - from:

Syntax: Delete from relation name;

example: Delete from std;

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b) Delete - From Where:

This is used to delete a selected from a relation

Syntax: Delete from relation_name
Where condition.

Example: Delete from student
where sno = 2

Conclusion:

Thus we have implemented DDL and DML commands along with integrity constraints.

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